

Assignment 3

K- Nearest Neighbours (KNN)

Problem Statement

An industrial machine operates under different temperature conditions and its fuel consumption varies accordingly. The following dataset shows the relationship between the machine's operating temperature and its fuel consumption.

Dataset

Operating Temperature ($^{\circ}\text{C}$)	Fuel Consumption (L/hr)
40	12
45	14
50	15
55	18
60	20
65	22
70	25
75	27

Given: $k = 3$

where $k \rightarrow$ no. of nearest neighbors

Using KNN Regression algorithm, manually predict the fuel consumption when the operating temperature is 58°C

i.e. $T = 58^{\circ}\text{C}$

Distance Calculation

Euclidean distance

$$\text{Distance} = \sqrt{(x_i - x)^2} \quad (\text{in Here}) \quad (\text{Since there is only 1 data})$$

where $x_i \rightarrow$ Temperatures of the given dataset

$x \rightarrow$ Given temperature i.e. 58°C

$$\text{For } 40^\circ\text{C} \quad \text{Distance} = \sqrt{(40 - 58)^2} = 18$$

$$\text{For } 45^\circ\text{C} \quad \text{Distance} = \sqrt{(45 - 58)^2} = 13$$

$$\text{For } 50^\circ\text{C} \quad \text{Distance} = \sqrt{(50 - 58)^2} = 8$$

$$\text{For } 55^\circ\text{C} \quad \text{Distance} = \sqrt{(55 - 58)^2} = 3$$

$$\text{For } 60^\circ\text{C} \quad \text{Distance} = \sqrt{(60 - 58)^2} = 2$$

$$\text{For } 65^\circ\text{C} \quad \text{Distance} = \sqrt{(65 - 58)^2} = 7$$

$$\text{For } 70^\circ\text{C} \quad \text{Distance} = \sqrt{(70 - 58)^2} = 12$$

$$\text{For } 75^\circ\text{C} \quad \text{Distance} = \sqrt{(75 - 58)^2} = 17$$

Nearest Neighbor Selection Table

Operating Temperature ($^\circ\text{C}$)	Fuel Consumption (L/hr)	Distance
40	12	18
45	14	13
50	15	8
55	18	3
60	20	2
65	22	7
70	25	12
75	27	17

Given $k=3$ i.e. 3 nearest neighbours are required

The ones with the least distance are the ~~neig~~ nearest neighbours.

Operating Temperature (°C)	Fuel consumption (L/hr.)	Distance
60	20	2
55	18	3
65	22	7

These are the 3 nearest neighbours obtained from the nearest neighbor selection table

Average

$$y = \frac{\text{Sum of fuel consumption of the } k \text{ nearest neighbors}}{k}$$

where $y \rightarrow$ Fuel consumption at 58°C temperature

$$y = \frac{20 + 18 + 22}{3}$$

$$y = 20 \text{ L/hr}$$

Therefore the Fuel consumption at Operating temperature of 58°C is 20 L/hr

File Edit Selection View Go Run Terminal Help

coding

EXPLORER

- CODING
 - New folder
 - aj.py
 - Assignment 2_PowerOutput.py
 - data.csv
 - KNN_Dataset.csv
 - KNN_FuelConsumption.py
 - LoadExtension_ALPHIN.py
 - LoadExtension.py
 - ml_program.py
 - power_data.csv

KNN_FuelConsumption.py > ...

```
1 import pandas as pd
2 from sklearn.neighbors import KNeighborsRegressor
3 import math
4 data = pd.read_csv("KNN_Dataset.csv")
5 x = data[["Temperature"]]
6 y = data[["Fuel_Consumption"]]
7
8
9 model = KNeighborsRegressor(n_neighbors=3)
10 model.fit(x,y)
11
12 temp_1 = 58
13 y_pred_1 = model.predict([[temp_1]])
14 print("Predicted Fuel Consumption for Temperature 58: ", y_pred_1)
15
16
17 temp = int(input("Enter the temperature: "))
18 y_pred = model.predict([[temp]])
19 print("Predicted Fuel Consumption: ", y_pred)
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python Python

```
PS D:\coding> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/coding/KNN_FuelConsumption.py
C:\Users\alphi\AppData\Local\Programs\Python\python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but KNeighborsRegressor was fitted with feature names
  warnings.warn(
Predicted Fuel Consumption for Temperature 58: [[20.]]
Enter the temperature: 97
C:\Users\alphi\AppData\Local\Programs\Python\python314\Lib\site-packages\sklearn\utils\validation.py:2691: UserWarning: X does not have valid feature names, but KNeighborsRegressor was fitted with feature names
  warnings.warn(
Predicted Fuel Consumption: [[24.66666667]]
PS D:\coding>
```

Tip: Try the Plan agent to research and plan before implementing changes.

+ KNN_FuelConsumption.py

Describe what to build

+ Auto

Local Default Approvals

Ln 12, Col 12 Spaces: 4 UTF-8 CRLF Python Python 3.14 (64-bit)